[SIN-WI-2428_A]Work Instruction: Swift Ray 1 BLE Provisioning with NRF Connect Mobile App

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Revision History

Revision	Changes	Author	Date
A	Initial Release	Trevor Folska-Fung	2022-11-16

Table of Contents

1. Introduction

This work instruction is to be used for serial number provisioning Sinatra devices after assembly via the nRF mobile application

1.1 Scope

This work instruction applies to all Ray 1 devices that will be functional to take images for both internal and external use.

1.2 Applicable and Reference Documents

Equipment/Supplies Required	Name of Tool	Description
SUP-C021	Devboard with connector cable (5-colour)	Used for Flashing Sinatra device (if not completed)
EQUIP-019	Iphone XR	Production phone used for provisioning serial number

1.3 Tools and Supplies Required

Document Title	Document Number
Sinatra Packaging & Labelling Component Specification	Sinatra Packaging Labelling Component Specification [SIN-CS-1987_B]
Sinatra Assembly Work Instruction	WI_ Sinatra Assembly Work Instruction (Hardware Electrical) [SIN-WI-2130_D]_
Prod Ops - Sinatra Inventory Assembled Devices (Device History Record)	Prod Ops - Sinatra Inventory Assembled Devices (Device History Record) [SIN-DHR-2152_B].xlsx

1.4 Training Required

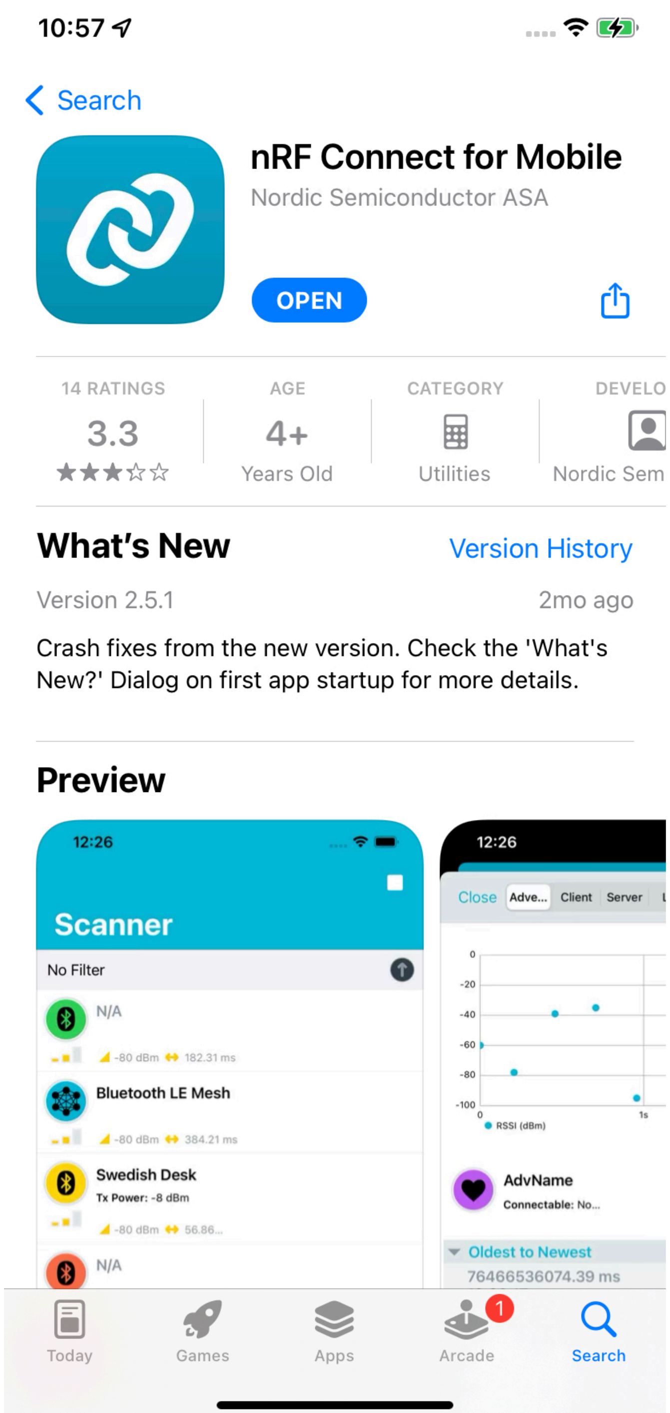
YES – Ray 1 Production Personnel

2. Work Instructions

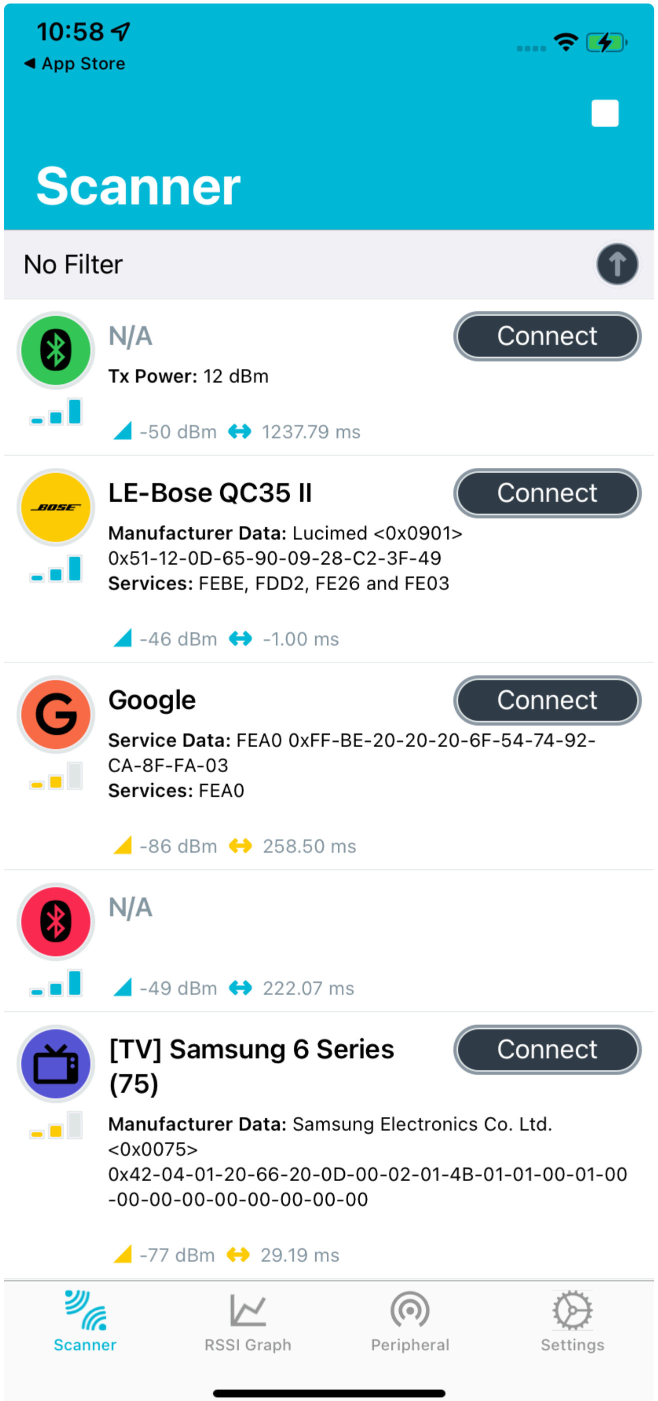
📘 Download and Install nRF Connect for Mobile App

nRF Connect can be run on both iPhone and Android.

1. Ensure the nrF Connect app is on your production phone, if not, download the nrF Connect app from the app store:

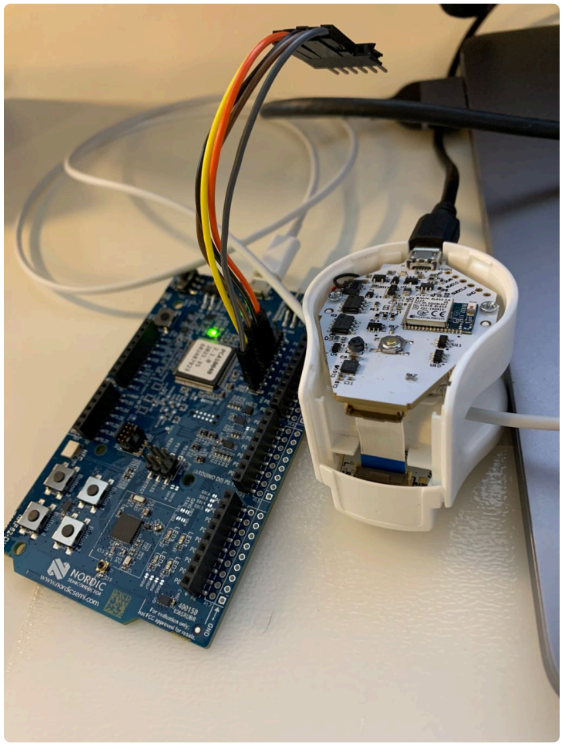


2. Once dowloaded, launch the app:

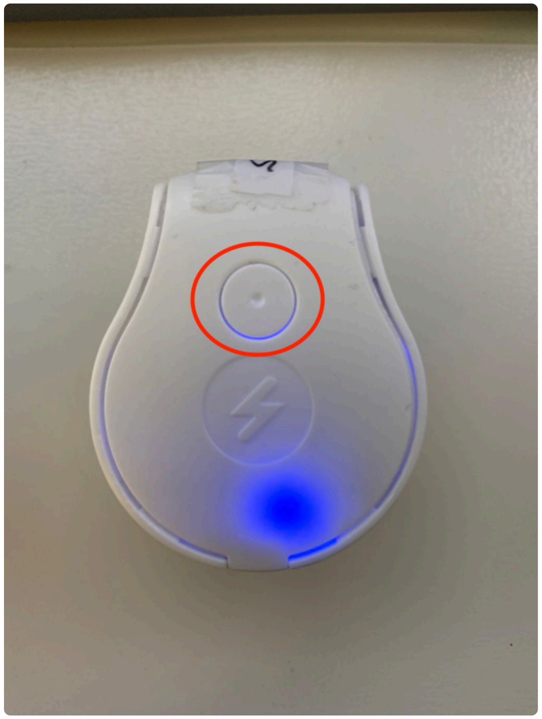

10:58
App Store
Scanner
No Filter
N/A
Connect
Tx Power: 12 dBm
-50 dBm 1237.79 ms
LE-Bose QC35 II
Connect
Manufacturer Data: Lucimed <0x0901>
0x51-12-0D-65-90-09-28-C2-3F-49
Services: FEBE, FDD2, FE26 and FE03
-46 dBm -1.00 ms
Google
Connect
Service Data: FEA0 0xFF-BE-20-20-20-6F-54-74-92-
CA-8F-FA-03
Services: FEA0
-86 dBm 258.50 ms
N/A
-49 dBm 222.07 ms
[TV] Samsung 6 Series
(75)
Connect
Manufacturer Data: Samsung Electronics Co. Ltd.
<0x0075>
0x42-04-01-20-66-20-0D-00-02-01-4B-01-01-00-01-00
-00-00-00-00-00-00-00-00
-77 dBm 29.19 ms
Scanner
RSSI Graph
Peripheral
Settings

📗 Steps to Burn Serial Number (Provision) into Ray 1

1. Follow above steps 1-20 in: [Work Instruction: Board-Flashing Procedure for Ray 1]and ensure flashing cable is removed from RAY1 device

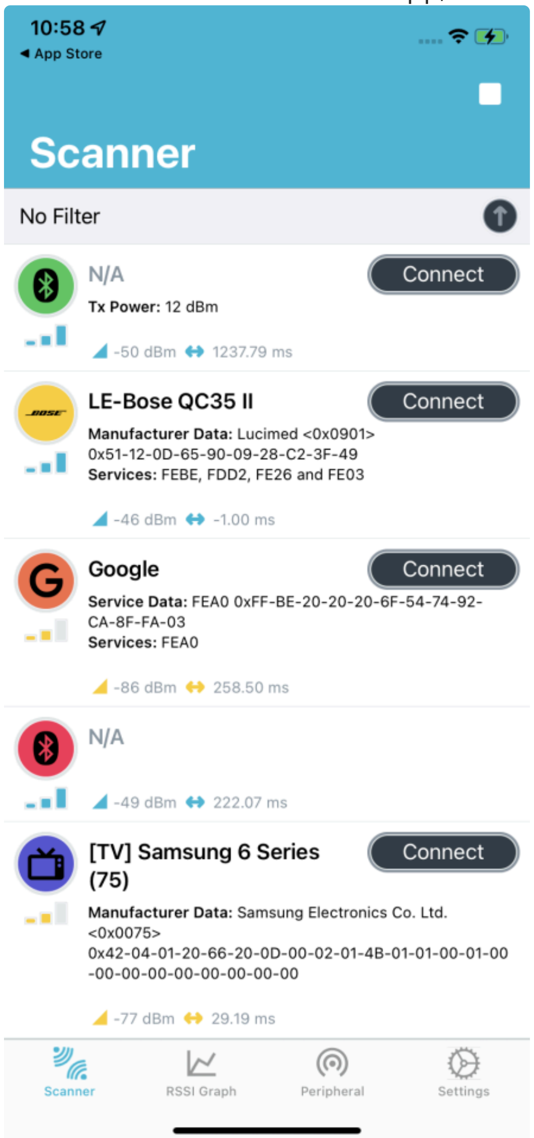


2. Press button to wake up Ray1:

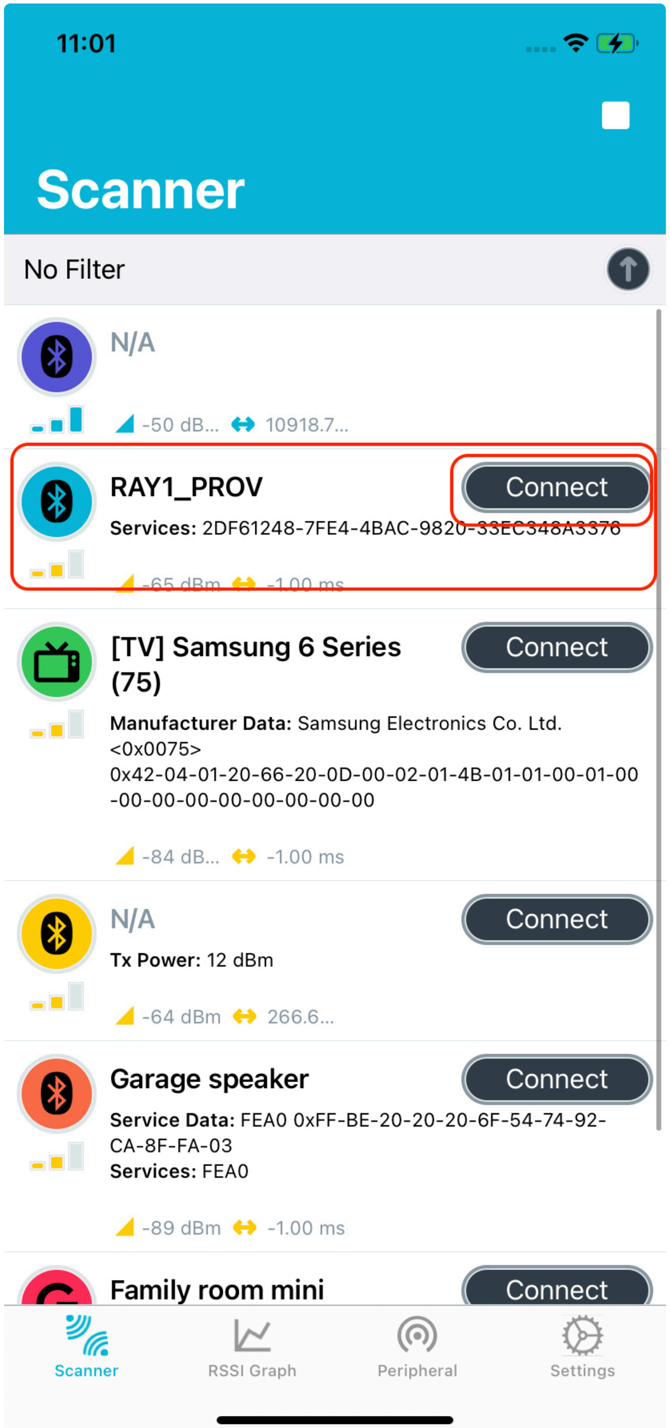


3. Ray1 will flash blue quickly signifying it is booted in provisioning mode:

4. In the nRF Connect for mobile app, view scanner screen:

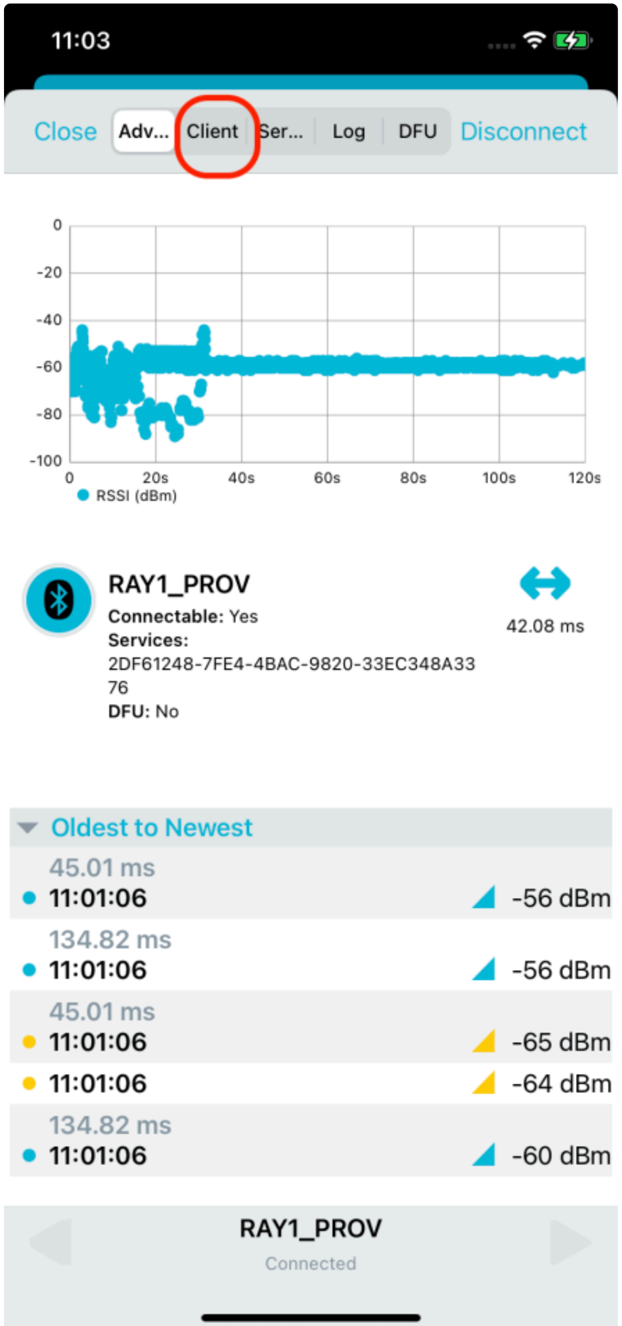


5. Connect to RAY1_PROV from the available bluetooth device list:

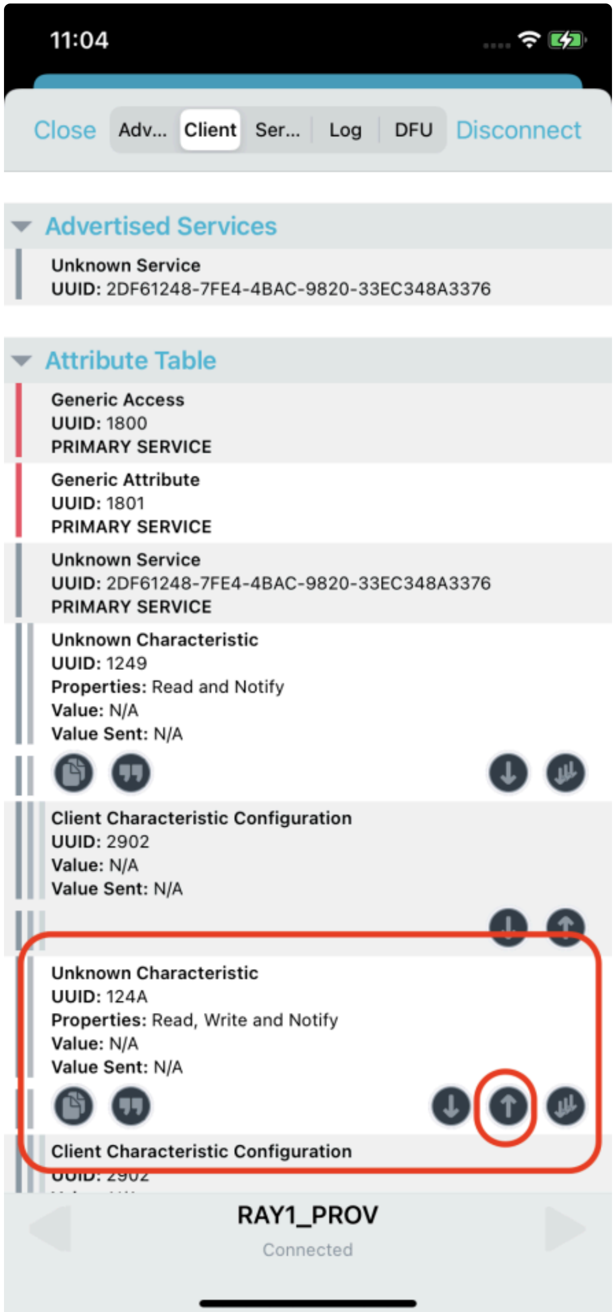

11:01
Scanner
No Filter
N/A
-50 dB... 10918.7...
RAY1_PROV
Connect
Services: 2DF61248-7FE4-4BAC-9820-33EC348A3376
-65 dBm -1.00 ms
[TV] Samsung 6 Series (75)
Connect
Manufacturer Data: Samsung Electronics Co. Ltd. <0x0075> 0x42-04-01-20-66-20-0D-00-02-01-4B-01-01-00-01-00-00-00-00-00-00-00-00-00
-84 dB... -1.00 ms
N/A
Connect
Tx Power: 12 dBm
-64 dBm 266.6...
Garage speaker
Connect
Service Data: FEA0 0xFF-BE-20-20-20-6F-54-74-92-CA-8F-FA-03
Services: FEA0
-89 dBm -1.00 ms
Family room mini
Connect
Scanner
RSSI Graph
Peripheral
Settings

6. If RAY1_PROV does not show up in scan try pressing RAY1_PROV button once to wake up RAY1_PROV device. Ray 1 device should turn on its 4 Blue LEDs once bluetooth is connected signifying device is ready for provisioning command:

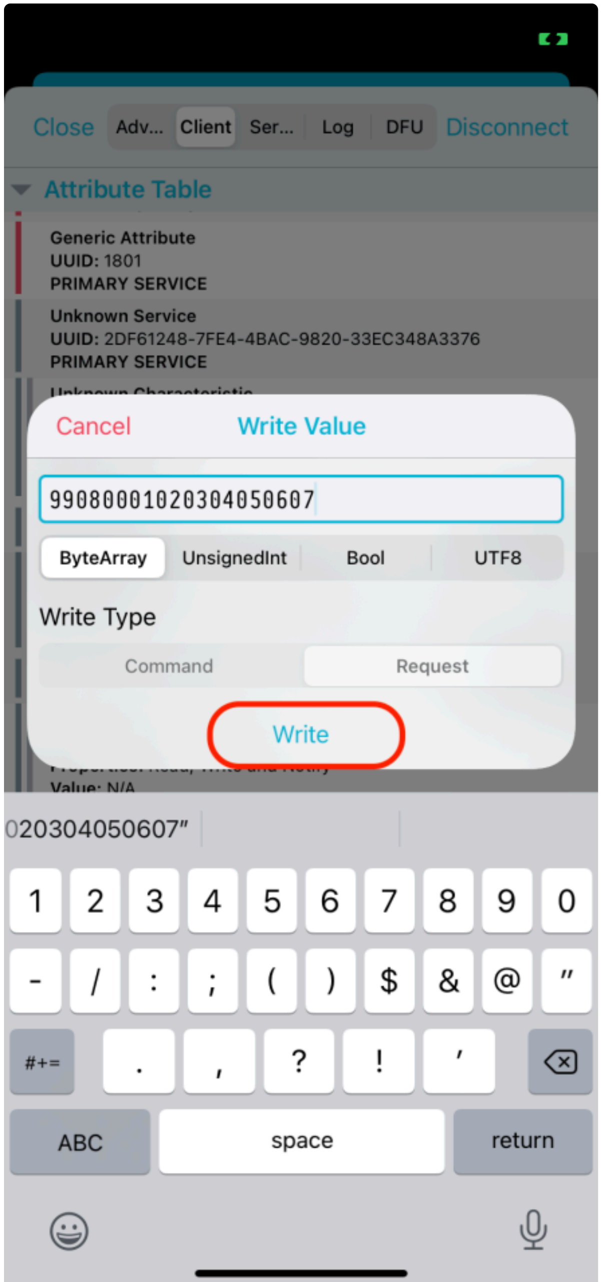
7. In the nRF Connect for mobile app, tap client once connected to RAY1_PROV:



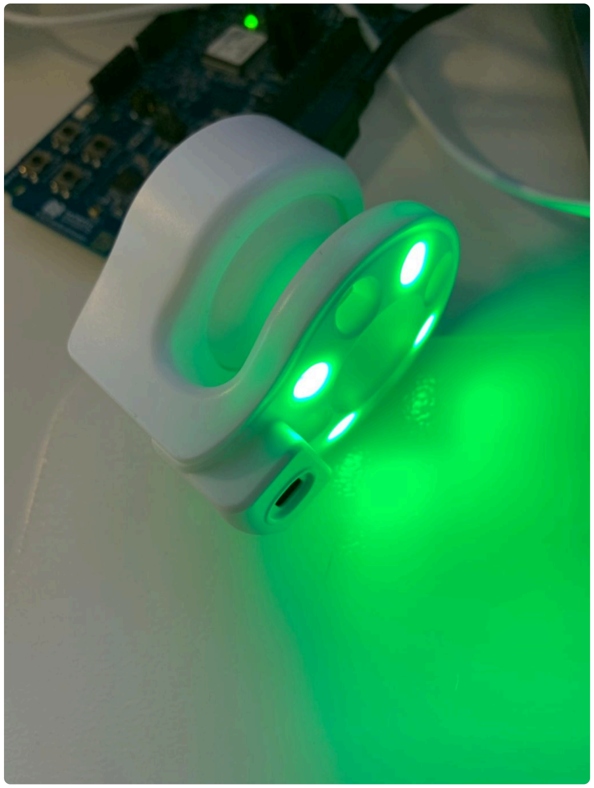
8. In the nRF Connect for mobile app, under service 124A, tap on the up arrow:



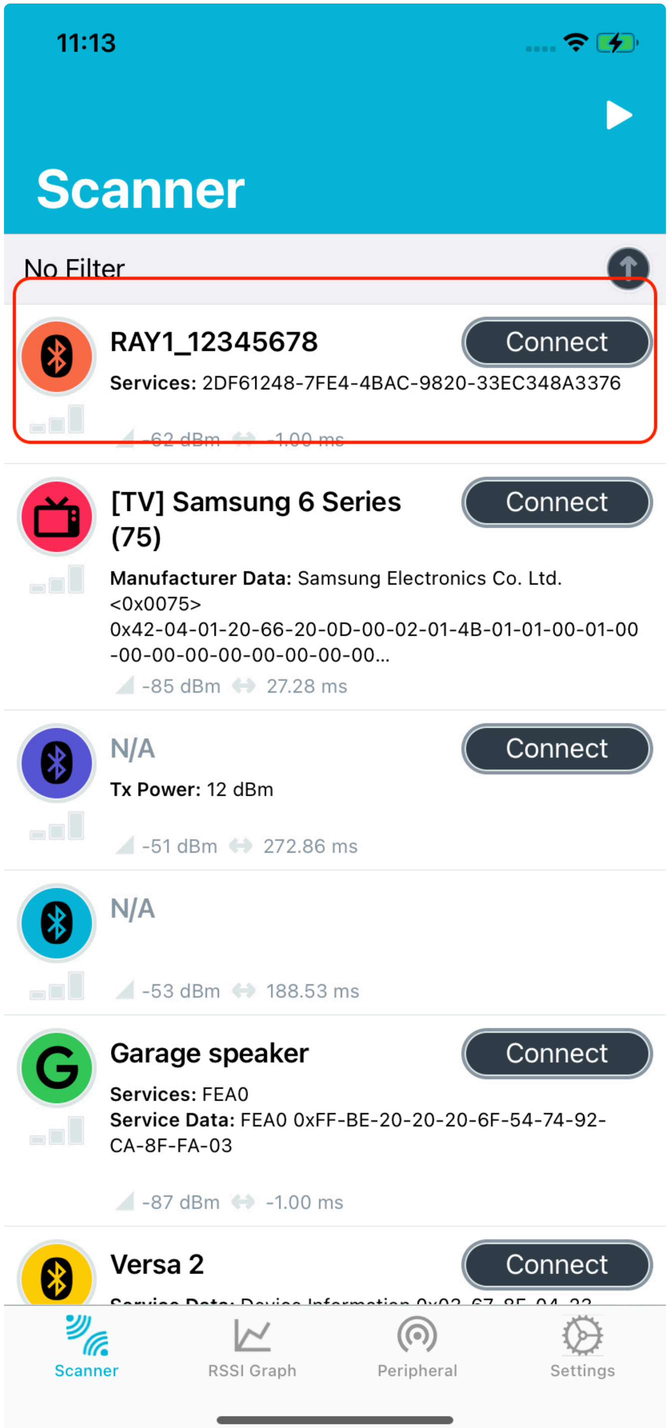
9. Enter the following byte array into the text box (Ex: 990800010203040506070) and tap write:



10. The Ray 1 device will flash its green to signify successful provision. If provisioning is unsuccessful it will flash purple:



11. A hard reset is now required (hold button for > 10 s). This will change device name (Ex: from RAY1_PROV to RAY1_12345678). To verify this device name change, in the nRF Connect for mobile app, view the new peripheral name:



12. Provisioning is complete and the Ray 1 device can be used as normal.

📋 Related articles

- BLE Provisioning with NRF Connect Desktop App [SIN-WI-2429_A] (Knowledge Base)
  - kb-how-to-article, devboard
- Flashing New Firmware with nRF52 Devboard (Knowledge Base)
  - kb-how-to-article, devboard
- Firmware Flashing Dev Kit Items (Knowledge Base)
  - devboard
- Firmware Flashing Dev Kit Items (Swift Ray)
  - dev-kit, devboard

Page Attachments